

Artemisia terpene diversity across scales: biosynthetic evolution and bounded antiparasitic evidence

Abstract

The genus *Artemisia* contains volatile monoterpenes and sesquiterpenes, less volatile diterpenes and triterpenes, and structurally diverse sesquiterpene lactones (STLs). These metabolites are often discussed as though a species name, a pathway gene, or an oil profile were sufficient to predict pharmacology. Here we present a source-auditable living scoping review that instead treats the specimen, accession, tissue, preparation, analytical method, parasite stage, and evidence level as the units of comparison. The current Terpedia package registers 535 sources, 540 source-linked evidence records, 125 antiparasitic records, 19 parasite–protein interaction records, 179 compound/specimen records, and 365 audited article claims; 50 sources and 59 linked evidence rows have completed full-text verification, while 56 queue sources remain open. We synthesize terpene classes with biosynthetic compartmentalization, regulatory and environmental variation, and a bounded comparative-genomics analysis. Artemisinin biosynthesis in *A. annua* is the best-resolved gene-to-metabolite benchmark, but it should not be used as a proxy for the genus. Malaria provides the strongest validated antiparasitic case: artemisinin has direct parasite-relevant mechanistic support, whereas non-artemisinin matrix and metabolomics studies mainly identify candidate contributors to preparation-level activity. Wormwood (*A. absinthium*) provides a separate vermifuge hypothesis supported by in-vitro nematode and limited animal evidence, but constituent attribution, host selectivity, and modern clinical translation remain unresolved. We propose a phylogeny-aware program linking voucher-authenticated metabolomics, secretory-tissue multi-omics, common reannotation, enzyme assays, and fractionated parasite testing. The result is a concise synthesis for journal submission with the complete evidence and provenance archive retained in Terpedia.

Keywords: *Artemisia*; terpene; sesquiterpene lactone; artemisinin; wormwood; malaria; vermifuge; chemotype; comparative genomics; Terpedia

1. Introduction

Artemisia is a chemically and ecologically heterogeneous genus. Volatile profiles can change with genotype, population, tissue, developmental stage, harvest, drying, storage, extraction, and analytical platform. Nonvolatile terpenoids are subject to an additional set of constraints: solvent selectivity, tissue localization, thermal instability, stereochemistry, ionization behavior, and structural dereplication. The appropriate biological unit is therefore not “the *Artemisia* terpene profile,” but a documented specimen or accession measured under a defined preparation and analytical context.

This distinction matters for antiparasitic interpretation. An essential oil is a mixture, an extract is a matrix, a purified compound is a defined exposure, and a transcript or homolog is a molecular hypothesis. These objects cannot be merged without losing the information needed to infer causality. The same problem appears in evolutionary genomics: a terpene-synthase (TPS) copy, an annotated cytochrome P450, or a shared orthogroup does not establish native product formation, pathway flux, or parasite activity.

Several important reviews already establish the breadth of the field. Essential-oil chemistry and anti-infective activity were synthesized at genus level by Abad et al. and updated for 2012–2017 by Pandey and Singh. Ivanescu et al. focused on STL analysis and biological activities; Trendafilova et al. reviewed edible *Artemisia* species and selected STLs; and Hussain et al. provided a broad review of traditional use, phytochemistry, pharmacology, and toxicology. Recent reviews of STL biosynthesis and evolution, *A. annua* production, antimalarial chemistry, and botanical anthelmintics further demonstrate that the literature is substantial rather than absent (Abad et al. 2012; Pandey and Singh 2017; Ivanescu et al. 2015; Trendafilova et al. 2021; Hussain et al. 2024).

The gap addressed here is not another catalogue of compounds. It is an evidence model connecting four scales: (i) compound identity and abundance; (ii) biosynthetic enzymes, regulation, and compartment; (iii) phylogeny, genome organization, and chemotype variation; and (iv) parasite phenotype, target evidence, host controls, and translation. Malaria and artemisinin are used as the high-confidence anchor. Wormwood is treated as a distinct vermifuge case because historical santonin chemistry, *A. absinthium* preparations, and modern veterinary nematode assays are related but not interchangeable.

2. Review scope and evidence framework

2.1 Living scoping-review design

The review was organized around five domains: compound occurrence and analytical identity; pathway, enzyme, and compartment evidence; genotype, population, tissue, developmental, and environmental variation; comparative genomics and gene-family evolution; and antiparasitic phenotype, target, and safety evidence. Searches combined *Artemisia* terms with terpene classes, analytical methods, biosynthesis, genomics, malaria, helminths, vectors, and safety terms. The frozen retrieval snapshot contains 4,765 unique PubMed records from seven query families, of which 1,387 were prioritized for title-level review and 1,323 had abstracts retrieved. A source-linked full-text eligibility queue covers 106 unique antiparasitic sources representing 125 evidence rows, prioritizing malaria and *A. absinthium* vermifuge records.

The package is deliberately a living scoping review rather than a completed systematic review. It contains 1,408 bounded title/abstract decisions, covering all 1,387 priority-queue records plus 21 historical or out-of-queue decisions. Full-text verification now covers 50 open-access or publisher-full-text antiparasitic sources and 59 linked evidence rows; 56 queue sources remain unverified, and no result is pooled as a final quantitative estimate. The PRISMA-ScR mapping is archived in `prisma-scr-checklist.md`; the search strings and retrieval counts are in `search-log.md` and `review-flow.md`. Reviews are used for discovery and context, while quantitative and mechanistic statements are preferentially anchored to primary studies.

Evidence was assigned according to what the source actually established. Structure-confirmed isolation or an authentic analytical standard supports compound identity in the tested material. A library-only GC-MS assignment is lower confidence. Heterologous enzyme activity supports catalytic capacity under assay conditions, not native flux. Transcript abundance, co-expression, and homology prioritize candidates but do not prove product identity. Parasite growth inhibition establishes a preparation-specific phenotype; causal target claims require biochemical, structural, genetic, or activity-based target engagement.

2.2 Chemical classes and evidence imbalance

Table 1. Chemical classes and minimum interpretation requirements.

Class	Representative <i>Artemisia</i> chemistry	Typical evidence	Minimum comparison fields
Monoterpenes (C10)	α -/ β -pinene, limonene, camphor, 1,8-cineole, borneol, thujones, artemisia ketone	GC-MS/GC-FID of oils or headspace; occasional isolation	Voucher, tissue, chemotype, stereochemistry, drying, retention index, standard
Sesquiterpenes (C15)	germacrene D, β -caryophyllene, α -humulene, davanone, spathulenol, caryophyllene oxide	Volatile profiling, isolation, enzyme assays	Preparation, abundance units, analytical confidence, tissue, stage
Diterpenes (C20)	species-specific diterpenoids and phytol-associated products	Solvent extraction, LC-MS, NMR, diTPS assays	Extraction polarity, structural confirmation, co-occurring matrix, native abundance
Triterpenes (C30)	amyriins, friedelin-, cycloartane- and sterol-associated chemistry	Solvent extraction, LC-MS, derivatized GC-MS, NMR, OSC assays	Surface/storage compartment, derivatization, standard, tissue and developmental context
Sesquiterpene lactones	artemisinin, arteannuin B, artemisinic acids, santonin, guaianolides, germacranolides	LC-MS/HPLC, NMR, isolation, enzyme and pathway studies	Stereochemistry, lactone structure, analytical standard, tissue, preparation, dose

Volatile mono- and sesquiterpenes dominate the searchable literature because routine GC methods make them accessible. Diterpenes, triterpenes, and STLs are less directly comparable because they require extraction, LC-MS, NMR, or targeted structural workflows. This is a sampling effect, not evidence that C10 and C15 chemistry is biologically more important. Conversely, a long list of isolated C20 or C30 compounds does not imply high abundance or antiparasitic relevance.

3. From precursor supply to chemical phenotype

3.1 Pathway architecture and compartment

The plastidial MEP and cytosolic mevalonate pathways supply IPP and DMAPP, which are condensed into prenyl diphosphates. GPPS-related activity supplies C10 substrates for monoterpene synthases, while FPPS supplies FPP for C15 sesquiterpene synthases. C20 and C30 chemistry depends on additional prenyltransferase and cyclase branches. TPS enzymes create hydrocarbon skeletons; P450s, reductases, dehydrogenases, oxidases, acyltransferases, glycosyltransferases, and spontaneous reactions generate the oxygenated and rearranged products detected in plants.

The pathway is spatially organized. Glandular secretory trichomes can combine precursor supply, TPS activity, oxidation, storage, and chemical defense in a restricted tissue. In *A. annua*, immunolocalization and trichome transcriptomics placed key artemisinin-pathway activities in secretory trichomes, while tissue-expression studies showed that terpene-associated transcripts vary among roots, stems, leaves, and reproductive tissues (Olsson et al. 2009; Wang et al. 2009; Olofsson et al. 2011). A metabolite detected in bulk leaf is therefore not necessarily produced uniformly across the leaf, and bulk expression can obscure the cells that control accumulation.

Secretory capacity is itself regulated. AaMIXTA1 perturbation altered glandular-trichome initiation, cuticle-related transcription, and artemisinin content; AaPDR3 affected β -caryophyllene accumulation in T-shaped trichomes; and promoter-reporter work showed age- and tissue-dependent *CYP71AV1* expression (Shi et al. 2018; Fu et al. 2017; Wang et al. 2013). These results support a model in which chemical evolution can act through cell type, transport, storage, and cuticle architecture in addition to enzyme sequence.

3.2 Artemisinin as a benchmark, not a genus-wide proxy

Artemisinin biosynthesis in *A. annua* provides the strongest gene-to-metabolite reference in the genus. ADS converts FPP to amorpha-4,11-diene; *CYP71AV1* oxidizes pathway intermediates; DBR2 and ALDH1 support the route toward dihydroartemisinic acid; and terminal chemistry includes nonenzymatic steps. Early cloning, recombinant assays, trichome-enriched expression, and isotope-feeding experiments established multiple pathway steps independently (Bouwmeester et al. 1999; Chang et al. 2000; Teoh et al. 2006; Zhang et al. 2008; Schramek et al. 2010).

The regulatory evidence is similarly layered. AaWRKY1, AaNAC1, AaORA, AaMYB108, jasmonate-linked ERF factors, cold-responsive AabHLH112, miR160-ARF1, and post-translational AaPP2C1-AaAPK1 regulation have each been connected to pathway expression or metabolite accumulation through perturbation, promoter, interaction, or expression experiments (Jiang et al. 2016; Lv et al. 2016; Lu et al. 2013; Xiang et al. 2019; Guo et al. 2023; Zhang et al. 2019). These studies establish causal regulation within tested *A. annua* backgrounds. They do not demonstrate conserved function in *A. absinthium*, *A. argyi*, *A. herba-alba*, or any other congener.

The same caution applies to production engineering. Recurrent selection increased artemisinin in *A. annua*, *CYP71AV1*/CPR co-expression increased production in transgenic lines, and a 2025 AaDHAADH variant improved dihydroartemisinic-acid production in engineered yeast (Paul et al. 2010; Shen et al. 2012; Guo et al. 2025). These are strong interventions for production, but increased plant titer is not equivalent to increased parasite selectivity, bioavailability, or clinical efficacy.

3.3 Evidence for other terpene classes

Monoterpene synthase assays show that closely related enzymes can generate distinct product spectra. In *A. annua*, AaTPS2, AaTPS5, and AaTPS6 differed in recombinant products and responded to wounding and hormone treatments (Ruan et al. 2016). *A. annua* seed isolation recovered sesquiterpenes, monoterpenes, and a diterpene, while trichome work functionally characterized OSC2 and *CYP716A14v2* in aerial-cuticle triterpenoid production (Brown et al. 2003; Moses et al. 2015). These records anchor C10, C15, C20, and C30 chemical space but do not define a genus-wide repertoire.

STLs are particularly important because the α -methylene- γ -lactone or related electrophilic functionality can react with nucleophilic protein residues. Yet structural class does not predict a single biological outcome. Guaianolides, germacranolides, eudesmanolides, artemisinin-type endoperoxides, and santonin-like compounds differ in stereochemistry, oxidation, ring fusion, reactivity, stability, and exposure. A structure-confirmed STL should therefore enter the evidence map first as a chemical occurrence; antiparasitic function requires a defined assay and, ideally, a matched host counter-screen.

4. Chemotype, tissue, environment, and analytical context

Across *Artemisia*, chemotype is an emergent phenotype rather than a fixed species label. Population, geography, phenology, plant age, light, temperature, drought, nutrient status, wounding, hormone treatment, and microbial association can alter both total production and the relative balance among branches. In *A. frigida* and *A. absinthium*, population and geography alter volatile composition; in *A. annua*, LED spectrum, plant density, soil copper, hydrogen sulfide, nitric oxide, and developmental stage alter artemisinin or volatile outputs (*A. frigida* variability study; Judzentiene et al. 2016; Ram et al. 1997; Zehra et al. 2020; Nomani et al. 2022; Wani et al. 2023). These effects are biologically informative but are not automatically transferable as cultivation recommendations or parasite interventions.

Postharvest chemistry is a major confounder in antiparasitic comparisons. Extended shade drying changed *A. monosperma* oil yield and composition, while hot-air drying/fixation in *A. argyi* reduced selected oxygenated monoterpenes and increased selected sesquiterpenes (Mohammed and Alfazi 2026; Ming et al. 2025). Wormwood profiles also change through evaporation and parent-to-daughter transformations during drying and storage (Blagojević et al. 2015). An assay using “dried wormwood” without temperature, duration, storage, oil yield, and chromatographic composition is therefore difficult to compare with an assay using fresh plant, tincture, or distilled oil.

Analytical identity is equally important. Four *A. herba-alba* oils showed variable enantiomeric proportions of α -thujone, β -thujone, and camphor when chiral methods were applied (Said Mel et al. 2016). In *A. persica*, structure-confirmed monoterpenoids and bisabolane sesquiterpenoids expanded known chemical space, but limited activity in the reported tests did not establish antiparasitic function (Umaraliev et al. 2025; Umaraliev et al. 2026). The comparative dataset therefore retains presence/absence, relative abundance, and analytical detectability as separate variables.

5. Comparative evolutionary genomics: what is supported

5.1 A bounded comparative analysis

The Terpedia package includes a completed full two-proteome OrthoFinder run for *A. annua* and *A. argyi*, restricted pathway-family mapping, and a bounded three-species diagnostic adding *A. tridentata*. The two-proteome run assigned 111,235 of 129,761 proteins to 28,696 orthogroups, including 25,039 groups containing both species and 7,086 single-copy groups. These results demonstrate computational feasibility and identify candidate shared and lineage-restricted families.

They do not constitute a genus-wide evolutionary analysis. A two-tip species tree is necessarily trivial. The three-species pathway diagnostic is candidate-focused and depends on heterogeneous assemblies and annotations. Copy-number comparisons are sensitive to ploidy, homeolog retention, assembly collapse, gene prediction, and annotation release. Absence from an annotation is therefore not biological absence. The strongest current conclusion is that *Artemisia* terpene evolution can be investigated with an orthology-aware workflow, not that comparative genomics has already explained antiparasitic diversity.

5.2 From orthogroups to testable gene-metabolite hypotheses

The useful evolutionary unit is a linked chain rather than a gene list:

species tree → gene tree/orthology → tissue expression → enzyme product
 → native metabolite → preparation exposure → parasite phenotype
 → host selectivity and target engagement

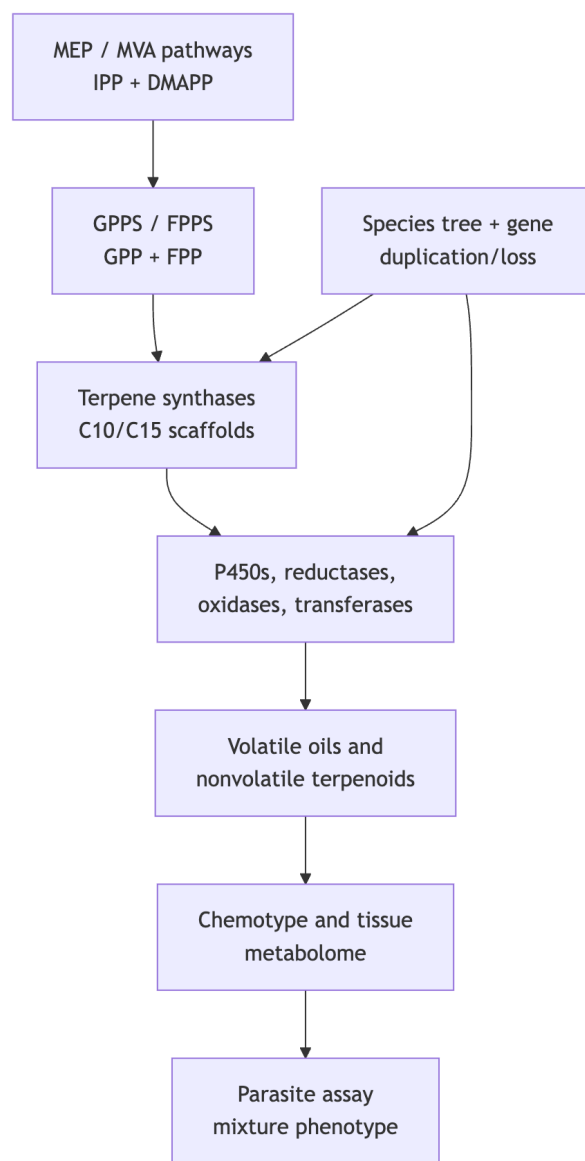
Each arrow is a potential failure point. A candidate TPS must be tested with a defined substrate and authentic product standards. A P450 candidate needs substrate-matched biochemical assays and tissue context. A metabolite association requires quantitative measurement in the same material used for the parasite assay. A parasite target hypothesis needs orthogonal confirmation rather than docking alone.

Table 2. Current comparative-genomics claims and the next discriminating experiment.

Claim level	Supported now	Not established	Discriminating experiment
Shared pathway ancestry	Candidate orthogroups and conserved pathway-associated sequences	Orthology across the genus and ancestral state	Common reannotation, species tree, gene-tree reconciliation
Gene-family expansion	Candidate TPS/P450/OSC differences in sampled genomes	Causal expansion or lineage-specific innovation	Chromosome-scale assemblies, synteny, homeolog-aware copy number
Catalytic divergence	Recombinant <i>Artemisia</i> enzyme and TPS evidence in selected cases	Native product and ecological function across species	Purified orthologs, isotope-labeled substrates, authentic standards
Chemotype evolution	Population, tissue, stage, and environmental associations	Heritable basis independent of environment	Common-garden multi-omics and QTL/allele-aware analysis
Antiparasitic evolution	Artemisinin pathway and defined assay anchors	Genus-wide gene-to-parasite causality	Matched chemotype, fractionation, target engagement, host counterscreens

The most productive next comparison is not “which species has more TPS genes?” It is whether pathway completion, tissue localization, and product spectra predict a defined antiparasitic phenotype after preparation is standardized. This is especially important for the distinction between *A. annua* artemisinin chemistry and *A. absinthium* thujone, STL, and historical santonin chemistry.

Figure 1. Conceptual route from precursor supply to phenotype.



6. Antiparasitic evidence

6.1 Evidence ladder and protein interactions

The interaction map separates direct target evidence, functional target-class evidence, enzyme or activity-based evidence, computational hypotheses, and unresolved targets. Artemisinin has the strongest direct malaria anchor, with heme activation, redox chemistry, proteome-wide or chemoproteomic signals, and parasite-state resistance studies contributing complementary rather than identical evidence. Calcium-homeostasis evidence extends to *Toxoplasma gondii* and related systems; recombinant enzyme assays for non-*Artemisia* STLs provide biochemical plausibility; docking against *Leishmania*, *Haemonchus*, or insect proteins remains hypothesis-generating unless paired with biochemical or genetic validation.

This separation prevents a common category error: treating a docking score as equivalent to target engagement, or treating an oil phenotype as evidence that one named constituent binds the proposed target. The 19-record interaction table is consequently a map of evidence gaps as much as a list of targets. For most non-artemisinin terpenes, the correct statement is that a preparation or purified compound has a phenotype and that target identity remains open.

To make this critical comparison reproducible, we applied the provisional non-compensatory rubric in evidence-appraisal-protocol.md to all 125 antiparasitic records. Twenty-one records had defined or purified chemistry, 36 had a profiled mixture, and 68 retained crude or unresolved attribution; 85 reported direct in-vitro parasite/vector assays, 37 included an in-vivo or organismal endpoint, and three were surrogate hemozoin assays. Explicit host/selectivity controls were present in 18 records, partial controls in 23, and absent or unresolved in 83. Only three records reached the E4/E5 functional-target-class level through the linked interaction map; 22 were hypothesis/stage-mechanism level and 100 were phenotype-only or unresolved. The overall appraisal therefore yielded 23 B-level characterized phenotypes and 102 C-level phenotypes with unresolved attribution; no record met the deliberately strict

A tier. This is a limitation of evidence resolution, not evidence that the preparations lack activity. Row-level judgments and their rule basis are archived in `evidence-appraisal.csv`.

6.2 Malaria: artemisinin as the validated anchor

Artemisinin is a sesquiterpene lactone with an endoperoxide bridge whose activation is central to its antimalarial pharmacology. The pathway and the parasite mechanism should be kept conceptually separate: a plant enzyme establishes production, while parasite experiments establish exposure, activation, damage, resistance, and selectivity. Artemisinin-related chemistry has been studied across asexual blood stages, gametocytes, resistant strains, rodent malaria, and whole-plant or tea preparations. These layers are not equivalent, but together they provide a much stronger evidence base than is available for most *Artemisia* terpenes.

The heme anchor is quantitatively reproducible in the full-text record. In synchronized 10–15-hour ring-stage *P. falciparum* TM267R, artemisinin IC₅₀ fell from 7.0 $\pm$ 1.1 nM to 3.1 $\pm$ 0.3 nM with 0.1 micromolar hemin and 1.9 $\pm$ 0.5 nM with 1 micromolar hemin. Severe-malaria plasma heme was 3.5 $\pm$ 0.7 micromolar versus 0.3 $\pm$ 0.2 micromolar in normal subjects; hemin increased TBARs in parasite systems, and vitamin E attenuated the oxidative signal. Quinoline comparators were not potentiated. This supports a purified-artemisinin heme/redox model in a synchronized blood-stage assay, not a sole parasite receptor, a crude-plant mechanism, or an infected-host treatment effect (Tangnitipong et al. 2012).

Whole-leaf and tea studies are especially informative about matrix effects and especially limited for constituent attribution. Mutant or chemically contrasted *A. annua* materials, tea infusions, and extract/isolated-artemisinin comparisons have reported activity that cannot always be predicted by artemisinin concentration alone. A gametocyte study compared *Artemisia* infusions with artemisinin, and a multistage study reported inhibition across parasite stages including relapse-relevant hypnozoite models (Balach et al. 2016; Bourdy et al. 2021; Cai et al. 2022). These results support matrix-aware pharmacology, not a general claim that crude tea is equivalent to an approved artemisinin combination treatment.

Recent metabolomics sharpens the matrix question. Shi et al. used UPLC-Q-TOF-MS/MS to compare wild and cultivated *A. annua*, reporting 24 flavonoids, 68 sesquiterpenoids, and 21 other compounds and nominating candidate non-artemisinin and arteannuin-B-related chemistry in a discovery and fractionation framework (Shi et al. 2022). The study identifies plausible contributors and prioritizes fractionation; it does not establish that a nominated sesquiterpene is the causal antimalarial molecule.

A full-text hydrotrope study provides a controlled matrix example. Cryoground leaves were extracted with optimized aqueous sodium- or cholinium-salicylate solutions, yielding 6.50 $\pm$ 0.10 and 6.44 $\pm$ 0.09 mg/g artemisinin. At matched 1 microgram/mL artemisinin-equivalent exposure against susceptible *P. falciparum* 3D7, the best extracts had IC₅₀ values of 0.0027 $\pm$ 0.0005 and 0.0034 $\pm$ 0.0009 microgram/mL versus 0.0066 $\pm$ 0.0025 microgram/mL for pure artemisinin; arteannuin B/J and several flavonoids co-occurred. The extracts were reported as nonhaemolytic at the closest IC₅₀ dose in cultured erythrocytes, but the single-line, mixed-chemistry design did not establish a causal terpene, in-vivo selectivity, or clinical safety (Ferreira et al. 2024).

Aubouy et al. compared teas from ten Benin plantations with French material using LC-HRMS and *P. falciparum* 3D7 assays. Artemisinin ranged from 0.3 to 15.7 mg/L, storage produced substantial losses in several Benin locations, and activity correlated positively with artemisinin while comparison with pure artemisinin suggested enhancing or limiting co-metabolites (Aubouy et al. 2025). This is strong origin-, storage-, and matrix-level evidence. It is not terpene-specific mechanism, standardized product performance, or clinical efficacy.

A full-text postharvest chemistry control further shows why a malaria-associated metabolomic feature cannot automatically be treated as an active terpene. Voucher-linked Beninese leaves and stems yielded the previously undescribed cadinapyridine sesquiterpene alkaloids annuanines A and B, with annuanine A confirmed by X-ray diffraction. Both compounds were inactive against chloroquine-resistant *P. falciparum* FcB1 at the reported concentrations and lacked notable activity in the tested Caco-2, VERO, and Thp1 cell models. Annuanine B was absent from fresh-plant tea and appeared after drying/storage; mixture experiments attributed activity to artemisinin rather than annuanine B. The proposed biosynthetic versus extraction/degradation origin remains unresolved, but the study provides a defined-compound negative control and a direct reason to include storage and artifact controls in untargeted *Artemisia* metabolomics (Fabre et al. 2025).

Full-text verification of two *A. afra* malaria studies sharpens the species boundary. A Cameroon acetone-extract comparison measured 19.3 mg/g artemisinin in dry *A. annua* extract, *A. afra* below the reported HPLC-UV detection limit, and LDH IC₅₀ values of 0.09 $\pm$ 0.01 versus 11.73 $\pm$ 2.15 microgram/mL. Late-stage *P. falciparum* metabolomics placed *A. annua* close to purified artemisinin, whereas *A. afra* shifted folate, carbohydrate, phospholipid, amino-acid, and lipid-precursor metabolism. A separate Senegal powder study found 93–98% day-3 activity for *A. annua* at 12.5–25 mg/kg artemisinin-equivalent doses but only 21–28% for single-dose *A. afra* and less than 5% after three daily high doses in *P. berghei* mice. Together these full-text controls support preparation- and host-dependent malaria interpretation; they do not identify a non-artemisinin terpene or falsify all tea-, origin-, or schedule-dependent *A. afra* hypotheses (Mamede et al. 2025; Walz et al. 2024).

The first full-text verification tranche makes these boundaries more concrete. Fresh-herb soaking or pounding changed artemisinin concentration from 14.5 mg/L in the dried-herb infusion to 293 mg/L in pounded juice, but only the pounded juice suppressed *P. berghei* parasitemia in the reported mouse comparison; the preparation effect therefore cannot be summarized as a universal tea

benefit (Wright et al. 2010). In a separate origin comparison, Cameroon and Luxembourg infusions differed in chromatographic profiles and beta-haematin inhibition, with the reported 10% dilution absorbance of 0.075 versus 1.515, respectively; this is a heme-surrogate and chemotype/matrix result rather than direct parasite efficacy (Akkawi et al. 2024). A voucher-linked *A. annua* study normalized several extract assays by artemisinin and compared sensitive and K13-mutant *P. falciparum* lines; activity remained strain- and exposure-time-dependent, and pooled malaria-exposed sera produced additional inhibition, supporting a host/matrix hypothesis without identifying a causal non-artemisinin terpene (Mouton et al. 2021). Finally, matched *A. annua* and artemisinin-free *A. afra* infusions inhibited blood and hepatic stages, including relapse-relevant models, but the active chemistry was not resolved to individual terpenes (Cai et al. 2022). These verified findings are archived row by row in `full-text-verification.csv`; they strengthen the evidence map while leaving constituent attribution and clinical translation open.

The newly verified 2013 comparison tested 16 geographically and temporally varied *A. annua* samples plus two *A. afra* samples as teas and chloroform extracts using HPLC-ELSD-normalized artemisinin measurements and a single chloroquine-sensitive *P. falciparum* 3D7 erythrocyte-stage assay. *A. annua* tea extracts averaged 0.75 microgram/mL extract IC50 and 4.97 ng/mL calculated artemisinin IC50 versus 5.48 ng/mL for pure artemisinin; chloroform extracts averaged 0.18 microgram/mL and 5.27 ng/mL, while *A. afra* teas were inactive above 25 microgram/mL and its chloroform extracts averaged 11.02 microgram/mL. The aged material, absent comprehensive terpene profiling, and single-stage design leave storage-sensitive constituents, prodrugs, and other life-cycle effects unresolved. The result supports artemisinin-dominant activity in the tested matrices, not the stronger genus-wide claim that artemisinin is the only active constituent (Mouton et al. 2013). The expanded verification set adds a more explicit plant-matrix pharmacology result. A petroleum-ether/silica extract quantified artemisinin, deoxyartemisinin, arteannuin I, dihydroartemisinic acid, and artemisinic acid as a sesquiterpene-rich mixture; at the lowest matched artemisinin dose, the extract produced approximately 42% murine inhibition versus approximately 4% for pure artemisinin, while rat exposure increased and Caco-2 assays implicated altered P-glycoprotein efflux. The study supports a preclinical matrix/host-transporter hypothesis, not a single-terpene parasite mechanism (Xu et al. 2025). A complementary genetic control found that removing major flavonoids while retaining wild-type artemisinin did not materially reduce *P. falciparum* Dd2 activity, whereas artemisinin-impaired lines lost activity; this strengthens artemisinin's role in that defined whole-leaf comparison without excluding matrix effects in other preparations (Czechowski et al. 2019). Silicon-induced trichome enlargement and higher infusion artemisinin likewise did not produce stronger *T. gondii* growth impairment, separating plant storage traits from parasite efficacy (Rostkowska et al. 2016).

Malaria resistance makes the evidence hierarchy more important. Artemisinin resistance is a parasite-state and genetic problem involving stress handling, proteostasis, heme detoxification, and redox homeostasis. A whole-plant or co-metabolite intervention could alter exposure or activity, but the responsible molecule and host selectivity must be established experimentally. The appropriate next study is a matched panel in which the same voucher-linked material is analyzed by targeted LC/GC methods, fractionated, tested against synchronized parasite stages and resistant lines, and paired with host-cell and heme/redox controls.

The newly verified malaria tranche includes both a resistance-evolution and a transmission-stage control. Orally delivered dried whole-plant *A. annua* reduced an artemisinin-resistant *P. yoelii* line more effectively than matched pure artemisinin and delayed stable resistance during serial *P. chabaudi* selection, but the study did not resolve a responsible terpene or establish a human regimen (Elfawal et al. 2015). In synchronized *P. falciparum* NF54 cultures, 5 g/L *A. annua* and *A. afra* teas reduced asexual parasites and gametocytes, with stronger effects tracking artemisinin content and only a weak artemisinin-deficient *A. afra* signal (Snider and Weathers 2021). These findings support a conditional artemisinin-centered, matrix-sensitive hypothesis, not generic terpene synergy or clinical transmission blocking.

Full-text verification adds a defined non-terpene resistance comparator. Infected male ICR mice received artemisinin plus the polymethoxylated flavonol chrysosplenetin at 40:80 mg/kg, with sensitive or resistant *P. berghei* K173 and Mdr1a wild-type or knockout backgrounds. Across the seven-day course, the combination retained 100% survival, approximately 30% inhibition, and approximately 20% parasitemia in resistant cohorts regardless of Mdr1a genotype, while tissue- and genotype-dependent ABC-transporter, ROS/GSH, cytokine, PI3K/AKT-mTOR, and MAPK responses changed. The study strengthens host/matrix and resistance-state hypotheses, but chrysosplenetin is not a terpene and no parasite-protein engagement or terpene-resolved botanical exposure was tested (Ji et al. 2025).

The following full-text checks add useful dose and mechanism boundaries. Powdered dried *A. annua* leaves containing 24 mg/kg artemisinin reduced *P. chabaudi* parasitemia more than equal pure artemisinin from 12 to 72 hours, but GC-MS-resolved leaves were not fractionated to identify a non-artemisinin terpene (Weathers et al. 2012). Purified DHA, artemether, and artesunate inhibited beta-hematin crystallization only under reduced conditions in a controlled heme-detoxification assay, while a chemical-genetic *P. falciparum* study linked DHA survival to UPR resolution and proteasome capacity across Kelch13 and proteasome-mutant backgrounds (Ribbiso et al. 2021; Rosenthal et al. 2024). These are mechanistic anchors for artemisinin drugs, not proof of a whole-plant terpene mechanism. Full-text verification of an Iranian three-species screen adds a surrogate-only boundary. Authenticated aerial parts of *A. ciniformis*, *A. biennis*, and *A. turanica* were successively extracted across five solvent systems; the strongest result was *A. ciniformis* dichloromethane extract in synthetic beta-hematin formation (IC50 0.92 +/- 0.01 mg/mL). The authors' lactone interpretation was not supported by terpene-resolved chemistry, and no parasite culture, host-selectivity, or in-vivo endpoint was included (Mojarrab et al. 2015).

Full-text verification also adds a comparative wormwood-family nematode control. Cold-methanolic *A. herba-alba* flower and

aerial-part extracts produced concentration- and time-dependent *H. contortus* adult-motility effects and 98.67% and 88.3% egg-hatch inhibition, respectively, at 1 mg/mL. Qualitative phytochemical screening—not terpene-resolved profiling—was used, and no mammalian selectivity endpoint was reported (Luka et al. 2020). This supports preparation-specific veterinary activity while leaving transfer to *A. absinthium*, thujone/STL attribution, in-vivo efficacy, and human safety unresolved. The newly verified goat coccidiosis study provides a negative translation control: a single 2 g/kg crude ethanolic *A. absinthium* dose was less effective than toltrazuril or amprolium, with oocysts persisting through day 28 (Iqbal et al. 2013). In a separate 46-plant tincture screen, 1.0% *A. absinthium* killed 85.5 +/- 16.5% of *Strongyloides papillosus* L1-2 larvae but only 19.4 +/- 6.7% of L3 larvae and none of *Haemonchus contortus* L3; no terpene profile or in-vivo endpoint was reported (Boyko and Brygadyrenko 2025). Full-text verification also adds a chemically resolved species control. Hydrodistilled flowering *A. absinthium* oil was dominated by myrtenyl acetate (47.10%), beta-thujone (11.41%), and pulegone (8.50%), whereas *A. dracunculus* oil was dominated by methyleugenol, beta-asarone, and elemicin. At 2 microliter/mL, *A. dracunculus* oil caused 100.0% and 94.9% mortality of *Meloidogyne incognita* and *M. chitwoodi* J2s at 72 hours, while *A. absinthium* caused only 11.4% and 7.5% mortality and increased egg hatching relative to control. sPLS and docking prioritized candidate constituents but did not establish causal molecules or nematode targets. This preparation- and species-specific result argues against a generalized wormwood vermifuge inference and is not direct evidence for gastrointestinal helminths (Evlice et al. 2026).

The next verification tranche includes a purified sesquiterpene-lactone transmission benchmark. Parthenin exposure eliminated *P. falciparum* mosquito oocysts at 1,000 ng/mL after stage V treatment and reduced median oocysts by 83% at 100 ng/mL; parthenolide retained a 47% reduction at 10 ng/mL. Parthenin also reduced exflagellation and blocked *P. berghei* ookinete maturation. Because these are non-*Artemisia* compounds, with stated parthenin cytotoxicity concerns and no mammalian therapeutic-window endpoint in the paper, the result motivates stage-resolved screening of authenticated *Artemisia* STLs but cannot be relabeled as wormwood activity (Balaich et al. 2016).

The full-text goat study supplies a particularly informative translation boundary. Market-purchased Peruvian leaves were oven-dried and extracted with 70% ethanol; the bulk extract caused 82.2% larval mortality and 81.79% motility inhibition against *Trichostrongylus* L3 at 200 mg/mL after 8 hours, yet 600 mg/kg per day for three days produced no significant fecal-egg-count reduction in five naturally infected goats. The material was not chemically resolved for thujone, santonin, or sesquiterpene lactones. This positive-in-vitro/negative-in-vivo pattern supports bioavailability, rumen-metabolism, dose, or preparation hypotheses—not a uniform claim that wormwood is either efficacious or inactive (Flores-Prado et al. 2025). A publisher-full-text Awassi-sheep study adds a contemporaneous positive veterinary result: a 95% ethanolic *A. absinthium* extract, prepared by repeated 16-hour maceration and given once at 1 g/kg, produced 66.79% FECR on day 7 and 76.77% on day 15 in six sheep, versus 100% for albendazole. Because the material had no lot-specific terpene fingerprint and the trial had 18 animals with short follow-up, this is a preparation-specific ovine signal rather than evidence of a causal thujone/STL mechanism or human safety (Irehan et al. 2026).

Two publisher-full-text checks add a defined preparation rationale and a C15 constituent lead. In the *Hymenolepis nana* study, authenticated but store-purchased Egyptian aerial parts were shade-dried and extracted in boiled water for 72 hours; the aqueous route was chosen to reduce thujone exposure. Oral extract at 400 and 800 mg/kg for three days reduced mouse EPG by 61.8% and 98.8% and worm burden by 56.5% and 98.5%, respectively, while praziquantel produced 100% reductions in the reported comparison. Adult worms exposed to 1 or 5 mg/mL showed dose-dependent paralysis, death, tegumental injury, lipid accumulation, nephridial-canal damage, and intrauterine-egg degeneration. The result is a strong crude-extract anticestodal signal with an explicit safety-motivated extraction choice, not a thujone/STL efficacy estimate or human vermifuge recommendation (Beshay 2018). In authenticated pre-flowering *A. cina*, bio-guided fractionation of a 3.86%-yield ethyl-acetate extract isolated the low-yield sesquiterpene cinic acid (0.01% of extract). The extract had LC50/LC90 values of 2.56/3.30 mg/mL against *H. contortus* L3, while the paper reports stronger cinic-acid activity in its isolated-compound assays; assay-dependent comparisons and the low yield require independent standardization and reconstitution. This provides a C15 constituent-to-helminth template, but not evidence that cinic acid occurs in *A. absinthium* or engages a defined parasite protein (Arango-De-la-Pava et al. 2024).

The full-text *A. cina* study provides a defined-mixture model for the wormwood hypothesis. Authenticated pre-flowering leaves and stems were macerated in ethyl acetate and fractionated by silica chromatography; GC-MS, HPLC-DAD, and multidimensional NMR identified peruvins as pseudoguaianolide sesquiterpene lactones and cinic acid as a related C15 sesquiterpene. In *H. contortus* L3 assays, peruvins plus 1-nonacosanol/hentriacontane produced 97.33% observed lethality versus 19.96% expected at matched 0.50LC25/0.50LC25 proportions (ratio 4.87), while peruvins plus cinic acid produced 40.42% versus 28.39% expected (ratio 1.42); the hydrocarbon/alcohol pair was additive for egg-hatching inhibition, and cinic acid alone showed no egg-hatching inhibition at the tested concentrations. The authors' protein-thiol and membrane-fluidity explanations remain hypotheses: there was no mammalian counterassay, infected-host test, exposure measurement, or direct parasite-protein experiment. This is therefore a strong comparative template for testing wormwood fractions, not evidence that peruvins or cinic acid occurs in *A. absinthium* (Arango-De-la-Pava et al. 2024).

An adjacent full-text *A. cina* study provides a useful non-terpene comparator. Bio-guided n-hexane fractionation of authenticated pre-flowering leaves and stems, followed by NMR and positive-mode ESI-MS, assigned the active C2F5 subfraction as a 63:37 mixture of 3'-demethoxy-6-O-demethylisoguaiacin and norisoguaiacin. The crude n-hexane extract caused 80 +/- 9% *H. contortus* L3 mortality at 4 mg/mL; active C1F5 and C2F5 fractions reached 100% mortality at 2 and 1 mg/mL, respectively. Because the

study tested a lignan mixture rather than purified individual standards, lacked terpene-resolved profiling, and reported no mammalian selectivity, infected-host, exposure, residue, or direct parasite-protein endpoint, it is retained as a fraction-to-phenotype comparator—not evidence for a terpene, thujone, santonin, or sesquiterpene-lactone mechanism (Higuera-Piedrahita et al. 2023).

6.3 Wormwood: vermifuge evidence without over-attribution

Wormwood's vermifuge history has multiple chemical and taxonomic layers. Santonin is a historically important anthelmintic STL associated with *Artemisia* taxa used for that purpose, but historical santonin exposure should not be treated as evidence that modern *A. absinthium* tea, oil, or powder has the same pharmacology. *A. absinthium* contains volatile thujones, camphor, 1,8-cineole, borneol, sesquiterpenes, STLs, phenolics, sterols, and other matrix constituents. A preparation-level helminth phenotype therefore cannot be assigned to “wormwood terpenes” without chemical matching and fractionation.

The modern evidence is real but heterogeneous. Methanolic *A. absinthium* was among the more active preparations in a multi-plant *Haemonchus contortus* egg and larval comparison, while UPLC-MS/MS documented selected secondary metabolites; the result supports preparation-level activity rather than a terpene-specific mechanism (Váradyová et al. 2018). An *Ascaris suum* study found concentration-dependent effects of alcoholic *A. absinthium* extract on embryogenesis and larvae, but the authors called for deeper phytochemical attribution (Băieș et al. 2022).

In experimentally infected lambs, wormwood extract showed strong in-vitro egg-hatch effects, yet group-level fecal egg counts did not differ significantly in the reported feeding trial (Mravčáková et al. 2020). A recent small Awassi-sheep study reported a 76.77% fecal egg-count reduction on day 15 for crude ethanolic extract compared with 100% for albendazole, but the 18-animal regional study requires independent chemical, dose, residue, and safety confirmation (Irehan et al. 2026). A low-input swine-farm study likewise reported antiparasitic effects after *A. absinthium* powder, but co-occurring interventions and complex powder chemistry prevent terpene attribution (Băieș et al. 2024).

Full-text verification of the lamb study confirms the central negative translation result: 24 lambs were assigned to four six-animal groups, wormwood stem was supplied at 1 g dry matter per day for 75 days, and the matched aqueous extract had *H. contortus* egg-hatch ED50/ED99 values of 1.40/3.76 mg/mL, yet treated-group fecal egg counts and terminal worm counts did not differ significantly from controls. The paper's chemical analysis emphasized phenolic fractions and reported 0.35 g/kg dry-matter total flavonoids; it did not provide a terpene-normalized exposure. A separate full-text *A. suum* study used a 70% ethanolic *A. absinthium* extract and detected sesquiterpene lactones among many metabolites, but reported only mixed-extract developmental inhibition. These records support a preparation-specific vermifuge hypothesis and a strong bioavailability/host-metabolism experiment, not a thujone- or lactone-specific claim (Mravčáková et al. 2020; Băieș et al. 2022).

A separate full-text *A. absinthium* schistosome study adds a useful constituent-control boundary. A dichloromethane leaf-rinse extract caused 100% mortality of adult *S. mansoni* at 200 microgram/mL, and fractionation yielded artemetin and hydroxypelenolide; both isolated compounds were inactive at 100 micromolar. The related-species comparator parthenolide from *Tanacetum parthenium* was active, but that result cannot be transferred to wormwood. The comparison demonstrates why “sesquiterpene lactone” is not itself an attribution: species, scaffold, preparation, concentration, and host counterassay must remain linked (Almeida et al. 2016).

Oil and volatile evidence broadens the hypothesis but does not resolve it. *A. absinthium* oil has shown trypanocidal or trichomonacidal activity in defined in-vitro systems, and thujone-rich *A. sieberi* oil has shown contact activity against poultry red mites; these are relevant antiparasitic phenotypes but are not equivalent to gastrointestinal nematode efficacy or human vermifuge use (Rivero-Cruz et al. 2015; Tabari et al. 2017). In *A. absinthium*, extraction-method comparisons against *Tetranychus urticae* showed different oil potencies and a candidate sesquiterpene association, but no single active molecule was established (Chiasson et al. 2001).

The best wormwood experiment is therefore not another crude-extract screen. It is a chemically fingerprinted, fractionated, dose-resolved assay against defined nematode life stages, with thujone and STL quantification, matched vehicle controls, mammalian cell or organoid counterscreens, and in-vivo pharmacokinetic and residue measurements. Only after that sequence could an evolutionary comparison test whether a *A. absinthium* chemotype or STL pathway predicts a reproducible vermifuge phenotype.

6.4 Other parasite systems as comparative controls

Other *Artemisia* evidence includes *Leishmania*, *Trypanosoma*, *Schistosoma*, *Eimeria*, *Toxoplasma*, ticks, mites, mosquitoes, and gastrointestinal helminths. These records are useful as comparative controls because they expose stage, route, host, and formulation effects. Purified STLs can show defined protozoan phenotypes; oils can show vector or ectoparasite effects; extracts can show rodent or ruminant activity. None should be pooled into one “antiparasitic potency” estimate without common units, matched exposure, parasite-stage annotation, and host controls.

Full-text extraction illustrates why oil composition and formulation must be retained together. In a vouchered Tunisian comparison, *A. campestris* oil was beta-pinene-rich and had a promastigote IC50 of 44 microgram/mL, whereas *A. herba-alba* oil was dominated by camphor, 1,8-cineole, chrysanthenone, and thujones and had an IC50 of 68 microgram/mL; macrophage CC50 values were 124.4 and 160 microgram/mL, respectively, in the study's matched counterassay (Aloui et al. 2016). These are oil-level,

promastigote-stage results rather than proof that beta-pinene, camphor, or thujone is causal. A formulated *A. absinthium* oil also produced intracellular-*L. amazonensis* activity, but the nanocochleate route and exposure cannot be generalized to wormwood vermifuge use (Tamargo et al. 2017).

The vector-control tranche is a separate translation category. Nanoliposomes containing oils from *A. annua*, *A. dracunculus*, and *A. sieberi* had species-dependent LC50 values against *Ae. aegypti* and *An. stephensi*, while *A. dracunculus* oil in a nanogel was more potent than its nanoemulsion against *An. stephensi* larvae; the studies characterized mixtures and formulations, not a malaria-parasite target or mammalian safety margin (Sanei-Dehkordi et al. 2023; Osanloo et al. 2022). Finally, antimalarial activity of *A. annua*-associated fungal endophyte extracts was traced to validated quinones such as emodin and physcion rather than plant terpenes, providing a useful negative attribution control (Alhadrami et al. 2021).

Negative or boundary records are particularly valuable. Some *Artemisia* preparations fail against a tested parasite, or show activity only in a formulation or one life stage. These observations help distinguish broad cytotoxicity from parasite selectivity and prevent publication bias from turning every reported plant effect into a positive mechanism.

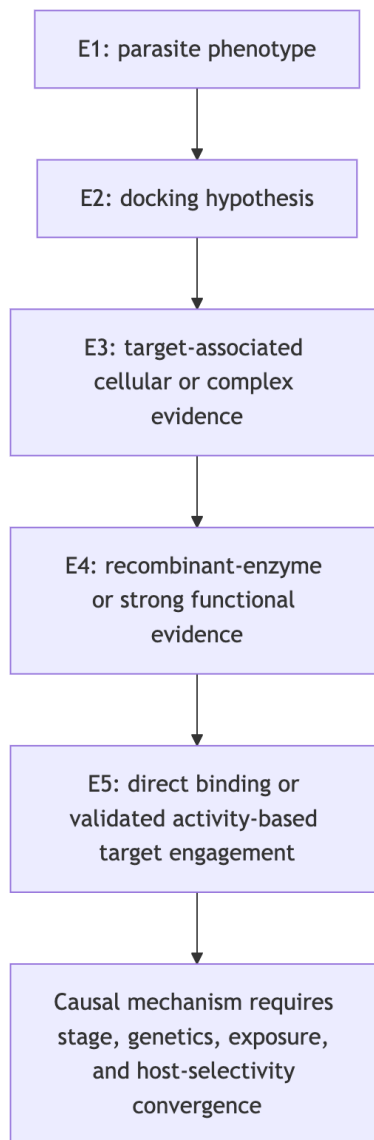
7. Safety and translation boundaries

Preparation identity is a safety variable. Artemisinin, an *A. annua* tea, a distilled essential oil, a hydroalcoholic extract, and *A. absinthium* powder are not interchangeable exposures. Thujone requires explicit quantification because neurotoxicity risk varies with chemotype, dose, route, and product. Regional *A. annua* and *A. afra* oils differ in camphor-, cineole-, and thujone-associated composition, while volatile pharmacokinetic studies show tissue distribution followed by elimination for defined oils rather than a universal safety profile (Radulović et al. 2013; Oyedele et al. 2009; Hou et al. 2021).

Host pharmacology is another translation gate. Artemisinin can affect host calcium-channel systems in experimental models, *Artemisia* teas can inhibit or induce drug-metabolizing enzymes depending on preparation, and reproductive studies in rodents raise preparation- and dose-specific concerns (Urban and Schaefer 2020; Kane et al. 2022; Abolaji et al. 2013). These findings do not define a human dose, but they show why parasite activity must be paired with host counterscreens, CYP/transporter testing, reproductive toxicology, and residue analysis.

The translation rule is simple: in-vitro activity is not a therapeutic dose; a mixture phenotype is not a single-constituent mechanism; a computational interaction is not target engagement; and increased plant production is not clinical efficacy. The evidence map retains these distinctions so that uncertainty is preserved rather than hidden in a pooled score.

Figure 2. Evidence ladder used for terpene–parasite target claims.



8. Testable hypotheses and research agenda

The integrated evidence supports six experimentally discriminating hypotheses.

1. Pathway completion will predict antiparasitic chemistry better than total TPS count. Across a phylogenetically sampled panel, complete precursor-to-oxidized-product modules should explain defined metabolite abundance better than raw TPS copy number. Test with common reannotation, gene-tree reconciliation, trichome expression, targeted metabolomics, and enzyme assays.
2. Secretory-tissue expression will outperform bulk-leaf expression. Cell-resolved or trichome-enriched transcriptomics should predict accumulated volatiles and STLs more accurately than bulk RNA-seq. Test with matched tissue sampling, spatial metabolomics, and isotope tracing.
3. Essential-oil antiparasitic activity will often be a chemotype-specific mixture property. Activity should be reproducible only when the full profile, stereochemistry, and exposure are matched. Test by reconstituting major and minor constituents at measured ratios and comparing whole oil, depleted oil, fractions, and purified compounds.
4. Artemisinin-like malaria phenotypes will depend on matrix and parasite state. Co-metabolites may alter uptake, metabolism, redox stress, or parasite-stage sensitivity without acting as standalone antimalarials. Test origin- and storage-matched teas against synchronized sensitive and resistant parasites, with LC-HRMS feature tracking and fractionation.
5. Wormwood vermifuge activity will require preparation-specific STL/volatile signatures. A reproducible nematode phenotype should track a chemical fingerprint and exposure window rather than the plant name alone. Test *A. absinthium* chemotypes against eggs, larvae, and adults, with thujone/STL quantification, host-cell counterscreens, and in-vivo residue studies.

6. Apparent gene-family expansion will be partly explained by cytotype and annotation. Homeolog retention, assembly collapse, and annotation choice should account for a measurable fraction of apparent TPS/P450/OSC expansion. Test chromosome-scale assemblies, allele-aware annotation, synteny, and reconciliation across a broad *Artemisia* species tree.

8.1 Minimum reporting standard

Future studies should report accepted taxon and voucher, accession or genotype, ploidy where known, tissue, developmental stage, locality, cultivation conditions, harvest and storage, preparation and extraction yield, analytical platform, authentic-standard or library status, stereochemistry, abundance units, parasite species and stage, inoculum, dose, endpoint, vehicle, positive control, host counter-screen, and raw-data availability. These fields are already represented across the Terpedia CSV schemas and should become the minimum unit for comparative synthesis.

9. Conclusions

There is sufficient primary research to support a strong *Artemisia* terpene review, but not to justify a single genus-wide potency score or a definitive evolutionary explanation of antiparasitic activity. The most secure conclusions are that *Artemisia* terpene chemistry is structurally broad and context-dependent; artemisinin biosynthesis and malaria pharmacology provide the best-resolved reference system; wormwood has a credible but preparation-specific vermifuge evidence stream; and comparative genomics currently provides testable prioritization rather than final causal explanation.

The central practical outcome is an evidence architecture. Chemical identity, tissue localization, pathway function, evolutionary history, parasite phenotype, target engagement, host selectivity, and translation must be linked without collapsing their uncertainty. Terpedia stores the full source registry, evidence matrix, screening records, interaction map, compound/specimen extraction, figures, code, checksums, and review-status record needed to extend this concise submission manuscript into a completed review.

Data and code availability

The complete auditable package is maintained in the Terpedia KB under the review-package directory and linked to the public *artemisia* source repository. The extended evidence-rich draft is retained as `article.md`; this file is the consolidated submission version. Large comparative-genomics inputs and outputs, search snapshots, structured evidence tables, claim audits, and SHA-256 manifests are versioned in the same package.